Proximal Supervised Fine-Tuning

接近监督微调

PROXIMAL SUPERVISED FINE-TUNING

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ABSTRACT

摘要

Supervised fine-tuning (SFT) of foundation models often leads to poor generalization, where prior capabilities deteriorate after tuning on new tasks or domains. Inspired by trust-region policy optimization (TRPO) and proximal policy optimization (PPO) in reinforcement learning (RL), we propose Proximal SFT (PSFT), a fine-tuning objective that incorporates the benefits of trust-region, ef-fectively constraining policy drift during SFT while maintaining competitive tun-ing. By viewing SFT as a special case of policy gradient methods with constant positive advantages, we derive PSFT that **stabilizes optimization** and **leads to generalization**, while **leaving room for further optimization in subsequent post-training stages**. Experiments across mathematical and human-value do-mains show that PSFT matches SFT in-domain, outperforms it in out-of-domain generalization, remains stable under prolonged training without causing entropy collapse, and provides a stronger foundation for the subsequent optimization.[1]

基础模型的监督微调（SFT）通常会导致泛化能力较差，即在对新任务或领域进行微调后，先前的能力会下降。受强化学习（RL）中信任区域策略优化（TRPO）和近端策略优化（PPO）的启发，我们提出了近端 SFT（PSFT），这是一种微调目标，能够结合信任区域的优势，在保持竞争力的同时有效限制 SFT 过程中的策略漂移。我们将 SFT 视为具有恒定正优势的策略梯度方法的一个特例，从而推导出 PSFT，它能够稳定优化并实现泛化，同时为后续的微调阶段留下进一步优化的空间。在数学和人类价值领域进行的实验表明，PSFT 在领域内与 SFT 表现相当，在领域外泛化方面优于 SFT，能够在长时间训练中保持稳定而不会导致熵坍缩，并为后续优化提供了更强的基础。

1 INTRODUCTION

1 引言

Recently, post-training has become a crucial part of the overall training process. In particular, reinforcement learning (RL) algorithms, such as PPO (Schulman et al., 2017) and GRPO (Shao et al., 2024), have demonstrated significant effectiveness when applied to language models (LMs) focused on reasoning tasks. As RL is scaled over time, foundation models gain the capacity to address com-plex problems through more profound and extended reasoning (OpenAI, 2024; Guo et al., 2025). These reasoning models offer an abundant and valuable latent thoughts (Ruan et al., 2025) across the internet. Numerous community efforts have focused on leveraging this knowledge via super-vised fine-tuning (SFT) (Guha et al., 2025; Li et al., 2025), a distillation approach valued for its efficiency and simplicity compared to RL.

最近，微调已成为整体训练过程中的关键部分。特别是强化学习（RL）算法，如 PPO（Schulman 等，2017）和 GRPO（Shao 等，2024），在应用于以推理任务为重点的语言模型（LMs）时表现出显著的效果。随着 RL 的不断扩展，基础模型获得了通过更深入和广泛的推理来解决复杂问题的能力（OpenAI，2024；Guo 等，2025）。这些推理模型在互联网上提供了丰富且有价值的潜在思维（Ruan 等，2025）。许多社区努力集中在通过监督微调（SFT）（Guha 等，2025；Li 等，2025）利用这些知识上，相比 RL，这种方法因其高效和简单而受到重视。

However, SFT models are often criticized for poor generalization (Huan et al., 2025). This limitation arises because SFT essentially performs behavior cloning, which can result in weak generalization when the fine-tuning dataset is suboptimal or distributionally misaligned with the pretraining data (Chu et al., 2025), potentially causing large policy updates (Schulman et al., 2015). RL fine-tuning (RFT) provides a promising alternative, as numerous studies have shown that RL better preserves the generalization capabilities that SFT tends to erode (Chu et al., 2025; Huan et al., 2025).

然而，SFT 模型经常因泛化能力差而受到批评（Huan 等，2025）。这种限制出现的原因是，SFT 本质上执行行为克隆，当微调数据集不够理想或与预训练数据分布不匹配时，可能导致泛化能力较弱（Chu 等，2025），从而引起较大的策略更新（Schulman 等，2015）。强化学习微调（RFT）提供了一种有前景的替代方案，因为大量研究表明，RL 更好地保持了 SFT 往往削弱的泛化能力（Chu 等，2025；Huan 等，2025）。

Another concern is maintaining the ability to explore. In practice, SFT is often used as a cold start to stabilize RFT training. The gains from RFT may largely come from refining capabilities already acquired during pretraining and SFT (Gandhi et al., 2025). However, excessive reliance on SFT can diminish a model's capacity for exploration (Xie et al., 2024), as it would cause the entropy collapse (Cui et al., 2025) and thereby constrain exploration during RL training (Yu et al., 2025).

另一个担忧是保持探索能力。实际上，SFT 通常被用作冷启动来稳定 RFT 训练。RFT 的收益可能在很大程度上来自于提升在预训练和 SFT 阶段已经获得的能力（Gandhi 等，2025）。然而，过度依赖 SFT 会削弱模型的探索能力（Xie 等，2024），因为它会导致熵坍缩（Cui 等，2025），从而在强化学习训练过程中限制探索（Yu 等，2025）。

Therefore, the current challenges lie in improving the **generalization** and **exploration keeping** of SFT models. To tackle these challenges, this paper proposes an improved fine-tuning strategy Proximal Supervised Fine-Tuning (PSFT) that avoids rote learning and achieves re-liable performance. Specifically, we establish a theoretical connection between SFT and RL, and

因此，当前的挑战在于提高 SFT 模型的泛化能力和探索保持能力。为了解决这些挑战，本文提出了一种改进的微调策略——邻近监督微调（PSFT），该策略避免了死记硬背，实现了可靠的表现。具体来说，我们建立了 SFT 与强化学习（RL）之间的理论联系，并

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[1]Code is available at https://github.com/zwhong714/PSFT.

代码可在 https://github.com/zwhong714/PSFT 获取。

introduce a novel objective based on a clipped surrogate objective that leverages the benefits of trust regions to constrain policy updates.

引入一种基于截断替代目标的新目标，利用信任区域的优势来约束策略更新。

Our experiments demonstrate that, compared to standard SFT, PSFT attains comparable performance on the target task while preserving the model's general capabilities. Moreover, PSFT mitigates entropy collapse during training and yields superior target and generalization performance in subsequent RL stages. We evaluate PSFT on both the mathematics and human value alignment domains, highlighting the broad applicability of PSFT in practice.

我们的实验表明，与标准的 SFT 相比，PSFT 在目标任务上取得了可比拟的性能，同时保留了模型的一般能力。此外，PSFT 在训练过程中缓解了熵坍缩问题，并在后续的强化学习阶段中表现出更优的目标性能和泛化性能。我们在数学和人类价值对齐领域评估了 PSFT，突显了 PSFT 在实际应用中的广泛适用性。

Our contributions are as follows:

我们的贡献如下：

- We propose PSFT, an optimization method for SFT that adopts a clipped surrogate objective, akin to PPO, to enforce trust-region-like constraints and thereby prevent excessive policy update. This approach maximally preserves the model's general capabilities while maintaining comparable performance on the target task.
- 我们提出 PSFT，这是一种针对 SFT 的优化方法，采用类似 PPO 的截断替代目标函数，以施加类似信任区域的约束，从而防止策略更新过度。这种方法在保持模型通用能力的同时，在目标任务上保持了相当的性能。
- PSFT effectively prevents entropy collapse and overfitting during SFT, thereby enabling subsequent RL stages to achieve more robust and superior results on both target-specific and general tasks.
- PSFT 有效防止了在 SFT 过程中的熵坍缩和过拟合，从而使后续的 RL 阶段在特定任务和通用任务上都能获得更加稳健和出色的结果。
- We extensively validate PSFT across various base models, tasks, and evaluation metrics, demonstrating its potential as a promising alternative to standard SFT.
- 我们在多种基础模型、任务和评估指标上广泛验证了 PSFT，展示了其作为标准 SFT 有前景的替代方案的潜力。

2 Preliminaries and Motivations

2 预备知识与动机

2.1 Markov Decision Process Formulation

2.1 马尔可夫决策过程表述

We formulate the problem of language modeling and optimization in the framework of a Markov Decision Process (MDP). An MDP is defined by a tuple (S, A, P), where: S is the state space, representing partial sequences. A is the action space, representing possible next tokens. $P(s' \mid s, a)$ is the transition probability of moving from state s to s' given action a. In the context of auto-regressive LMs, a query sequence $x := (x_1, \ldots, x_m) \in X$ initializes the state. A response sequence $y := (y_1, \ldots, y_n) \in Y$ is generated by sequentially sampling actions (tokens) from the policy π_θ. The joint probability of y given x is: $\pi_\theta(y \mid x) = \prod_{t=1}^{n} \pi_\theta(y_t \mid y_{<t}, x)$, where $y_{<t} := \{y_1, \ldots, y_{t-1}\}$ and the state at time t is $s_t = (x, y_{<t})$.

我们将语言建模和优化问题建模为马尔可夫决策过程（MDP）框架下的问题。一个 MDP 由一个元组(S, A, P)定义，其中：S 是状态空间，表示部分序列；A 是动作空间，表示可能的下一个标记；P(s′ |s, a)是从状态 s 在采取动作 a 后转移到状态 s′ 的概率。在自回归语言模型的背景下，一个查询序列 x := (x1, ..., xm) ∈ X 初始化状态。一个响应序列 y := (y1, ..., yn) ∈ Y 是通过依次从策略 π θ 中采样动作（标记）生成的。给定 x 时，y 的联合概率为： π θ (y|x) = ∏_{t=1}^n π θ (yt | y<t, x)，其中 y<t := {y1, ..., yt−1}，并且在时间 t 的状态 st = (x, y<t)。

2.2 SFT AS THE SPECIAL POLICY UPDATE

2.2 SFT 作为特殊政策更新

Supervised Fine-tuning. The training objective is to minimize the cross-entropy loss between the model's predicted token distribution and the ground truth tokens. Formally, the loss is defined as:

监督微调。训练目标是使模型预测的标记分布与真实标记之间的交叉熵损失最小化。正式地，损失定义为：

$$L_{\mathrm{SFT}}(\theta) = -\hat{\mathbb{E}}_{(s_t, a^*_t) \sim D} \left[\log \pi_\theta(a^*_t \mid s_t) \right], \tag{1}$$

LSFT(θ) = −E_{(s_t,a*_t)~D} [log π_ θ (a*_t| s_t)] , (1)

where (s_t, a^*_t) pairs are sampled from an offline dataset D.

其中 (st, a*t) 对是从离线数据集 D 中采样的。

Policy Gradient. In contrast, policy gradient directly samples trajectories from the current policy π_θ interacting with the environment. Using the policy gradient theorem, the corresponding loss function can be defined as follows:

$$L_{\mathrm{PG}}(\theta) = \hat{\mathbb{E}}_{(s_t, a_t) \sim \pi_\theta} \left[\log \pi_\theta(a_t \mid s_t) \hat{A}_t \right], \tag{2}$$

策略梯度。相比之下，策略梯度直接从当前策略 π θ 与环境交互中采样轨迹。利用策略梯度定理，相应的损失函数可以定义如下：

LPG(θ) = ˆE(st,at)~ π θ log π θ (at | st) ˆAt , (2)

where $\hat{A}_t$ is the estimated advantage function at step t.

其中 $\hat{A}_t$ 是第 t 步估计的优势函数。

From this perspective, SFT can be seen as a special case of policy gradient where sampling is from a fixed offline dataset D and the advantage is fixed as $\hat{A}_t = 1$ for the ground-truth action, which corresponds to maximizing the likelihood of expert tokens.

从这个角度来看，SFT 可以被视为策略梯度的一个特例，其中采样来自固定的离线数据集 D，优势固定为 ˆAt = 1，对应于最大化专家标记的似然。

2.3 PROXIMAL POLICY OPTIMIZATION

2.3 接近策略优化

Trust Region Policy Optimization. TRPO (Schulman et al., 2015) maximizes a surrogate objective by leveraging trajectories collected under the previous policy $\pi_{\theta_{\text{old}}}$, introducing importance sampling to reweight these samples for evaluating the new policy π_θ:

信任区域策略优化。TRPO（Schulman 等，2015）通过利用在旧策略 π θ old 下收集的轨迹来最大化一个替代目标，引入重要性采样来重新加权这些样本，以评估新策略 π θ：

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$$L(\theta) = \hat{\mathbb{E}}_{(s_t,a_t)\sim\pi_{\theta_{\text{old}}}} \left[\frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)} \hat{A}_t \right] = \hat{\mathbb{E}}_t \left[r_t(\theta) \hat{A}_t \right], \quad (3)$$

L(θ) = E_{(st,at)~ π _ θ ^{old}} \left[\log \frac{ π _ θ (at | st)}{ π _ θ ^{old} (at | st)} \right] \cdot \hat{A}_t = E_{\hat{t}} \left[\hat{r}_t(θ) \hat{A}_t \right], (3)

where $r_t(\theta)$ is the importance sampling ratio, and the superscript *CPI* refers to conservative policy iteration (Kakade & Langford, 2002). By introducing a trust region constraint on the KL divergence between π_θ and $\pi_{\theta_{\text{old}}}$, TRPO ensures that each policy update does not deviate too far from the reference policy.

其中，rt(θ) 是重要性采样比率，上标 CPI 表示保守策略迭代（Kakade & Langford, 2002）。通过在策略 π θ 与 π θ old 之间的 KL 散度上引入信任区域约束，TRPO 确保每次策略更新不会偏离参考策略太远。

Proximal Policy Optimization. However, directly maximizing L^{CPI} with a hard KL constraint, as in TRPO, can be difficult to optimize in practice. To address this, PPO (Schulman et al., 2017) modifies the surrogate objective by penalizing policy changes that move $r_t(\theta)$ too far from 1. The clipped surrogate objective is:

邻近策略优化。然而，像 TRPO 那样直接以硬 KL 约束最大化 LCPI，在实践中可能难以优化。为了解决这个问题，PPO（Schulman 等，2017）通过惩罚使 rt(θ)偏离 1 太远的策略变化来修改替代目标。裁剪后的替代目标为：

$$L^{\text{CLIP}}(\theta) = \hat{\mathbb{E}}_{(s_t,a_t)\sim\pi_{\theta_{\text{old}}}} \left[\min \left(r_t(\theta) \hat{A}_t, \ \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon) \hat{A}_t \right) \right]. \quad (4)$$

This clipping mechanism effectively defines a soft trust region: $r_t(\theta)$ is restricted to remain within a small neighborhood of 1, which prevents destructive updates. Compared to TRPO, PPO achieves similar stability and efficiency.

这种裁剪机制有效地定义了一个软信任区域：rt(θ) 被限制在 1 的一个小邻域内，从而防止了破坏性更新。与 TRPO 相比，PPO 在稳定性和效率方面达到了类似的效果。

3 Proximal Supervised Fine-Tuning

3 近端监督微调

Proximal Supervised Fine-Tuning aims to improve performance on the target SFT task while preserving general capabilities, preventing entropy collapse, and allowing further optimization.

邻近监督微调旨在在保持通用能力、防止熵坍缩的同时提升目标 SFT 任务的性能，并允许进一步优化。

3.1 From PPO to PSFT

3.1 从 PPO 到 PSFT

We revisit the TRPO and PPO objectives in RL. Both methods rely on importance sampling ratios $r_t(\theta)$ to reweight returns, while constraining these ratios to prevent the new policy π_θ from deviating excessively from the old policy $\pi_{\theta_{\text{old}}}$. Translating this idea to the supervised setting, where all actions are assumed to be "correct" (i.e., $\hat{A}_t > 0$), we simplify the advantage to $\hat{A}_t = 1$, and define the Proximal SFT loss as:

$$L_{\text{PSFT}}(\theta) = \mathbb{E}_{(s_t,a_t)\sim D} \min \left(\frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}, \text{clip}\left(\frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}, 1-\epsilon, 1+\epsilon \right) \right). \quad (5)$$

我们重新审视强化学习（RL）中的 TRPO 和 PPO 目标。这两种方法都依赖于重要性采样比率 rt(θ)来重新加权回报，同时约束这些比率，以防止新策略 π θ 与旧策略 π θ old 发生过度偏离。将这一思想转化为监督学习的设定（即假设所有动作都是“正确的”（即ˆAt > 0）），我们将优势函数简化为ˆAt = 1，并定义了近端 SFT 损失如下：

$$\text{LPSFT}(\theta) = \mathbb{E}_\{(st,at)\sim D\} \min \left(\frac{\pi\ \theta\ (at \mid st)}{\pi\ \theta\ \text{old}\ (at \mid st)}, \text{clip}\left(\frac{\pi\ \theta\ (at \mid st)}{\pi\ \theta\ \text{old}\ (at \mid st)}, 1-\epsilon, 1+\epsilon \right) \right). \quad (5)$$

This objective regularizes supervised learning updates by limiting the ratio between the new and old policy probabilities. We allow the old policy $\pi_{\theta_{\text{old}}}$—rather than the fixed initial model—to evolve dynamically. Intuitively, it discourages overconfident changes in token probabilities, thereby pre-serving existing capabilities. When changing to the online setting, this method is equivalent to standard SFT training.

这一目标通过限制新策略概率与旧策略概率之间的比率，对监督学习的更新进行正则化。我们允许旧策略 π θ old 动态演化，而不是固定初始模型。直觉上，这会抑制对 token 概率过于自信的更改，从而保留现有的能力。当转换到在线设置时，该方法等同于标准的 SFT 训练。

Warm-Up. Note that (s_t, a_t) in Eq. 5 is sampled from the offline dataset D. In the initial steps, $r_t(\theta)$ may yield a biased expectation since $\pi_{\theta_{\text{old}}}$ is not aligned with the distribution of D. The clipping mechanism constrains policy updates within an acceptable range, which may indirectly reduce the negative impact of this mismatch. A warm-up SFT phase on D can be introduced to better align the initial policy $\pi_{\theta_{\text{old}}}$ with the offline dataset, further improving the SFT/PSFT in-domain performance.

预热。请注意，式 5 中的(st, at)是从离线数据集 D 中采样的。在初始步骤中，由于 π θ old 与 D 的分布不一致，rt(θ)可能会产生有偏的期望。裁剪机制将策略更新限制在可接受的范围内，这可能会间接减少这种不匹配的负面影响。可以在 D 上引入一个预热的 SFT 阶段，以更好地将初始策略 π θ old 与离线数据集对齐，从而进一步提高 SFT/PSFT 在域内的性能。

3.2 Gradient Analysis

3.2 梯度分析

The gradient updated is confined to the trust region to perform fine-tuning, similar to that in PPO. The gradient is as follows:

梯度更新被限制在信任区域内以进行微调，类似于 PPO 中的做法。梯度如下：

$$\nabla_\theta L_{\text{PSFT}}(\theta) = \mathbb{E}_{(s_t,a_t)\sim D} \left[r_t \cdot \mathbb{I}_{\text{trust}}(r_t) \cdot \nabla_\theta \log \pi_\theta(a_t \mid s_t) \right],$$

∇ θ LPSFT(θ) = E_{(st,at)~D} [rt ・ Itrust (rt) ・ ∇ θ log π θ (at | st)] ,

0 $r_t > 1 + \epsilon$, (6) where $I_{trust}(r_t) =$ 1 otherwise.

0 rt > 1 + ϵ, (6) 其中 Itrust(rt) = 1 否则。

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The gradient analysis shows that if the train dataset distribution deviates significantly from the model distribution, these tokens will have no gradient, thereby avoiding large policy updates and maintain-ing generalization. Typically, the optimal value of ϵ is set to 0.2 or 0.28. A larger ϵ can lead to large gradients, as shown in Equation 6. Further analysis is provided in Section 5.2.

梯度分析表明，如果训练数据集的分布与模型分布有显著差异，这些标记将不会产生梯度，从而避免大的策略更新，保持泛化能力。通常，ε 的最佳值设置为 0.2 或 0.28。较大的 ε 会导致较大的梯度，如公式 6 所示。进一步的分析见第 5.2 节。

4 EXPERIMENTS

4 实验

In Section 4.1, we explore the training dynamics of our PSFT with standard SFT, and then evaluate both in-domain and out-of-domain performances of models fine-tuned with each method, focusing primarily on the mathematics domain. In Section 4.2, we attempt to verify the effectiveness of PSFT as the cold start point of RL. In Section 4.3, we further conduct experiments in the alignment domain to assess the universal applicability of our approach.

在第 4.1 节中，我们探讨了我们的 PSFT 与标准 SFT 的训练动态，并评估了使用这两种方法微调模型在领域内和领域外的性能，主要集中在数学领域。在第 4.2 节中，我们尝试验证 PSFT 作为强化学习冷启动点的有效性。在第 4.3 节中，我们进一步在对齐领域开展实验，以评估我们的方法的普遍适用性。

4.1 MAIN EXPERIMENTS ON MATH REASONING IN THE SFT STAGE

4.1 SFT 阶段中数学推理的主要实验

Setup. *(1) Models and Datasets.* We evaluate our method based on Qwen2.5-7B-Instruct (Yang et al., 2025) and Llama3.1-8B-Instruct (Dubey et al., 2024). Our training data focuses on the math domain, aiming to improve general LLM capabilities through math reasoning. We employ the OpenR1-Math-8192[2]long chain-of-thought (CoT) dataset (Face, 2025). *(2) Baseline.* We consider the SFT method and an SFT variant, denoted as SFT_{KL}, which incorporates KL divergence constraints into the loss function. The KL regularization coefficient is set to 0.5, while the ϵ parameter in PSFT is fixed at 0.28. More details can be found in Appendix A.1.1.

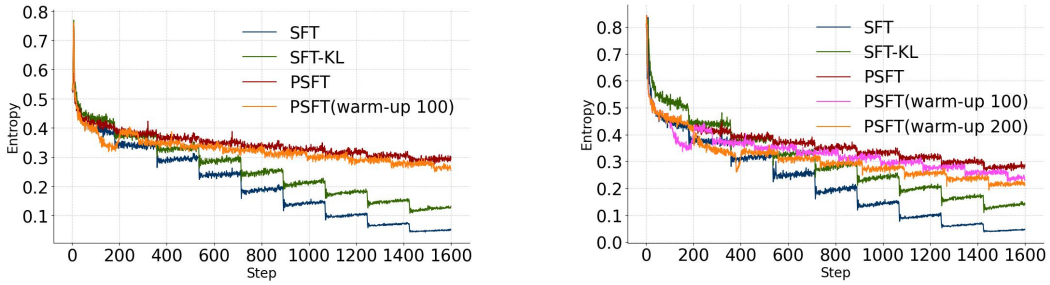
设置。(1) 模型和数据集。我们在 Qwen2.5-7B-Instruct (Yang 等人，2025) 和 Llama3.1-8B-Instruct (Dubey 等人，2024) 上评估我们的方法。我们的训练数据聚焦于数学领域，旨在通过数学推理提升通用大语言模型的能力。我们采用 OpenR1-Math-81922long 链式思维（Chain-of-Thought，CoT）数据集（Face，2025）。(2) 基线。我们考虑了 SFT 方法及其一种变体 SFTKL，该变体将 KL 散度约束纳入损失函数中。KL 正则化系数设置为 0.5，而 PSFT 中的 ϵ 参数固定为 0.28。更多细节请参见附录 A.1.1。

4.1.1 TRAINING DYNAMICS

4.1.1 训练动态

We begin by examining the training dynamics of each method. Specifically, we train each for around 10 epochs and plot the corresponding changes in entropy and performance. In this section, we use AIME-24 avg@32 to measure in-domain performance and GPQA (Rein et al., 2024) avg@8 to assess out-of-domain performance.

我们首先考察每种方法的训练动态。具体来说，我们对每种方法进行大约 10 个周期的训练，并绘制熵和性能的变化情况。在本节中，我们使用 AIME-24 avg@32 来衡量领域内性能，并使用 GPQA（Rein 等，2024）avg@8 来评估领域外性能。



(a) Qwen2.5-7B-Instruct (b) Llama3.1-8B-Instruct

(a) Qwen2.5-7B-Instruct (b) Llama3.1-8B-Instruct

Figure 1: Training dynamics of Entropy. Each 178 steps is one epoch.

图 1：熵的训练动态。每 178 步为一个周期。

Observation 1: PSFT avoids entropy collapse. The entropy evolution is shown in Figure 1. Compared to SFT and SFT-KL, PSFT produces a smoother entropy curve. For SFT and SFT-KL, entropy exhibits a marked decline after each epoch, indicating potential overfitting. This suggests that PSFT is capable of sustaining long-term, token-level fine-grained training without triggering entropy collapse. Notably, PSFT with a warm-up phase demonstrates the same stability, and the number of warm-up rounds plays a critical role in shaping the overall entropy level. Further evidence is provided in Figure 6a.

观察 1：PSFT 避免了熵坍缩。熵的演化情况如图 1 所示。与 SFT 和 SFT-KL 相比，PSFT 产生的熵曲线更加平滑。对于 SFT 和 SFT-KL 来说，在每个训练周期后熵都会明显下降，这表明可能存在过拟合现象。这表明 PSFT 能够在不触发熵坍缩的情况下维持长期的、基于 token 的精细训练。值得注意的是，带有预热阶段的 PSFT 表现出相同的稳定性，预热轮次的数量在塑造整体熵水平方面起着关键作用。进一步的证据见图 6a。

Analysis 1: PSFT matches or surpasses in-domain performance of standard SFT at a similar entropy level. The in-domain evaluation performance evolution is shown in Figure 2. PSFT

分析 1：PSFT 在相似熵水平下，其在域内的性能表现与标准 SFT 相当或更优。域内评估性能的演变如图 2 所示。PSFT

[2]The dataset undergoes filtering through Math-Verify to exclude instances with unverifiable answers or responses exceeding 8,192 tokens in length.

数据集通过 Math-Verify 进行过滤，以排除无法验证答案或响应长度超过 8,192 个标记的实例。

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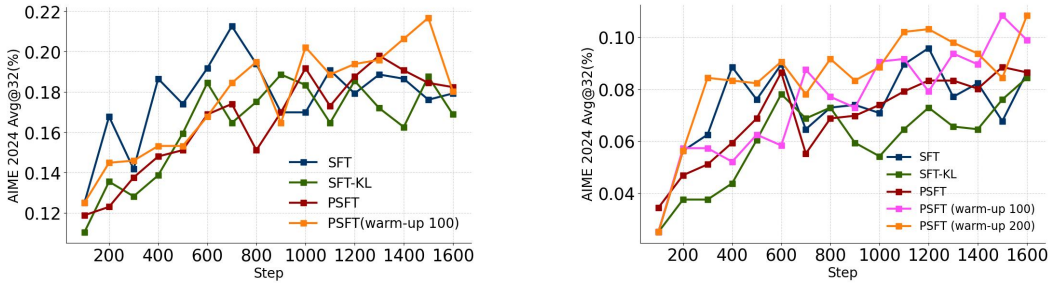
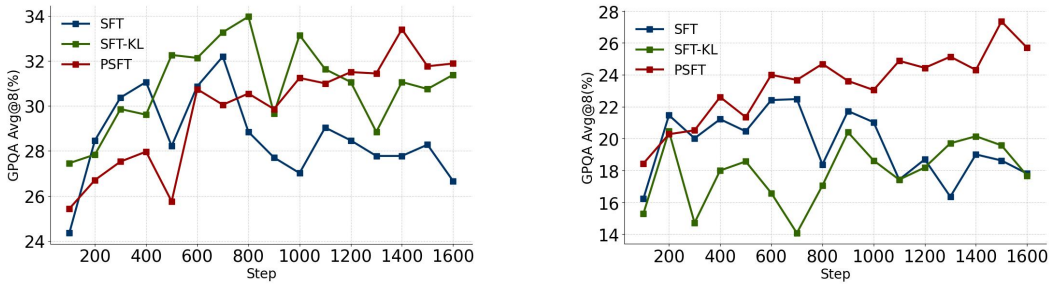


(a) Qwen2.5-7B-Instruct (b) Llama3.1-8B-Instruct

Figure 2: Training dynamics of in-domain performance.

(a) Qwen2.5-7B-Instruct (b) Llama3.1-8B-Instruct 图 2：领域内性能的训练动态。



(a) Qwen2.5-7B-Instruct (b) Llama3.1-8B-Instruct

(a) Qwen2.5-7B-Instruct (b) Llama3.1-8B-Instruct

Figure 3: Training dynamics of out-of-domain performance.

图 3：领域外性能的训练动态。

achieves comparable performance on targeted math tasks and can even outperform SFT under the same entropy level. For instance, at step 1300, PSFT attains an AIME24 score of nearly 20 with an entropy around 0.3. This entropy level corresponds to that of SFT between epochs 300 and 520, yet PSFT achieves a higher AIME score than SFT within this range. While the KL-based approach intuitively addresses distribution shift, considering both entropy trends and in-domain performance, it remains less effective than PSFT.

在针对数学任务上取得了可比的性能，甚至在相同熵水平下还能优于 SFT。例如，在第 1300 步时，PSFT 的 AIME24 得分接近 20，熵值约为 0.3。这一熵值对应于 SFT 在第 300 到 520 个周期之间的水平，但在此范围内，PSFT 的 AIME 得分高于 SFT。虽然基于 KL 的方法直观地解决了分布偏移问题，考虑了熵趋势和领域内性能，但其效果仍不如 PSFT。

Analysis 2: PSFT with warm-up operation surpasses standard SFT. Figure 2 illustrates the impact of introducing a warm-up phase into PSFT. We observe that the in-domain performance in-

creases steadily. Across both models, PSFT with warm-up consistently outperforms vanilla PSFT as well as the standard SFT baselines. Moreover, increasing the number of warm-up steps further en-hances in-domain performance. **This offers an effective way to fully exploit high-quality datasets** such as s1k (Muennighoff et al., 2025) and LIMO (Ye et al., 2025) while avoiding entropy collapse.

分析 2：带有预热阶段的 PSFT 优于标准的 SFT。图 2 展示了将预热阶段引入 PSFT 的影响。我们观察到，领域内性能稳步提升。对于两种模型而言，带有预热阶段的 PSFT 始终优于标准的 PSFT 以及标准的 SFT 基线。此外，增加预热步骤的数量进一步提高了领域内性能。这提供了一种有效的方法，可以充分利用高质量数据集，如 s1k（Muennighoff 等，2025）和 LIMO（Ye 等，2025），同时避免熵坍缩。

Analysis 3: PSFT can largely maintain the generalization of the model. The evolution of out-of-domain evaluation performance is shown in Figure 3. Increasing the number of training steps leads to a decline in generalization for SFT models. For example, after 700 steps, the SFT method on Qwen2.5-7B-Instruct exhibits a clear decline in performance. While SFT-KL alleviates this degra-dation to a certain degree, PSFT demonstrates substantially better generalization.

分析 3：PSFT 可以在很大程度上保持模型的泛化能力。图 3 展示了领域外评估性能的演变情况。增加训练步数会导致 SFT 模型的泛化能力下降。例如，在 700 步之后，Qwen2.5-7B-Instruct 上的 SFT 方法表现出明显的性能下降。虽然 SFT-KL 在一定程度上缓解了这种退化，但 PSFT 表现出显著更好的泛化能力。

Overall, benefiting from its robust entropy control during training, **PSFT avoids large fluctuations in policy updates.** As observed in both in-domain and out-of-domain evaluations, PSFT generally maintains an upward performance trend. In contrast, SFT and SFT-KL exhibit pronounced fluctua-tions in out-of-domain performance.

总体而言，得益于其在训练过程中强大的熵控制，PSFT 避免了策略更新中的大幅波动。在域内和域外评估中观察到，PSFT 通常保持性能的上升趋势。相比之下，SFT 和 SFT-KL 在域外性能上表现出明显的波动。

4.1.2 Detailed Evaluations

4.1.2 详细评估

Setup. *(1) Model setup:* For a fair comparison, **we select checkpoints based on in-domain per-formance.** For instance, on Qwen2.5-7B-Instruct, we use the SFT model trained for 700 steps, the SFT-KL model trained for 900 steps, and the PSFT model trained for 1300 steps. *(2) Evaluation*

设置。(1) 模型设置：为了公平比较，我们根据领域内性能选择检查点。例如，在 Qwen2.5-7B-Instruct 上，我们使用训练了 700 步的 SFT 模型、训练了 900 步的 SFT-KL 模型以及训练了 1300 步的 PSFT 模型。(2) 评估

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Table 1: Detailed results of **in-domain** performance. For AIME and AMC, the results are avg.@32. For the rest, the results are avg.@8.

表 1：领域内性能的详细结果。对于 AIME 和 AMC，结果为 avg.@32。对于其余项目，结果为 avg.@8。

Method	AIME24	AIME25	AMC	MATH-500	OlympidBench	Minerva	Avg.
			Qwen2.5-7B-Instruct				
Original	11.25	8.75	52.58	75.05	39.72	40.53	37.98
SFT	22.08	23.02	62.73	84.10	52.35	**43.66**	47.99
SFT_{KL}	19.27	21.56	**63.20**	83.55	**52.70**	42.19	47.08
$PSFT_{warm\text{-}up}$	**22.92**	**23.02**	62.42	**84.68**	52.30	43.38	**48.17**
PSFT	19.38	21.98	62.34	83.35	51.50	43.33	46.98
			Llama3.1-8B-Instruct				
Original	3.96	0.73	24.45	48.55	17.09	25.87	20.11
SFT	10.63	16.77	47.19	72.60	41.43	32.17	36.80
SFT_{KL}	9.58	16.15	45.55	69.83	39.19	26.75	34.51
$PSFT_{warm\text{-}up}$	**12.08**	**18.75**	**49.45**	**74.15**	**42.07**	**33.64**	**38.36**
PSFT	10.31	14.48	46.80	71.98	39.61	32.40	35.93

Table 2: Detailed results of **out-of-domain** performance. For GPQA, ARC-C, TruthfulQA, and IFEval, the results are avg.@8. For the rest, the results are pass.@1.

表 2：领域外性能的详细结果。对于 GPQA、ARC-C、TruthfulQA 和 IFEval，结果为 avg.@8。对于其余项目，结果为 pass.@1。

Method	GPQA	ARC-C	TruthfulQA	MMLU-Pro	SuperGPQA	HeadQA	$IFEval_{loose}$	Avg.
			Qwen2.5-7B-Instruct					
Original	31.38	91.54	66.10	54.99	27.59	73.41	**73.94**	59.85
SFT	32.89	92.22	63.14	58.98	**29.02**	74.65	54.42	57.90
SFT_{KL}	32.95	91.86	61.31	58.35	27.69	74.07	55.44	57.38
$PSFT_{warm\text{-}up}$	**33.27**	92.09	66.37	**59.28**	28.37	75.24	55.07	58.53
PSFT	33.21	**92.29**	**67.16**	59.18	28.10	**75.82**	73.03	**61.26**
			LLama3.1-8B-Instruct					
Original	24.62	80.96	55.14	43.70	18.16	68.20	**78.10**	52.70
SFT	19.38	87.73	67.08	50.29	21.95	73.52	33.45	50.49
SFT_{KL}	18.18	87.02	67.03	47.02	19.81	71.95	34.19	49.31
$PSFT_{warm\text{-}up}$	23.99	**89.72**	68.55	56.03	25.26	77.71	33.08	53.48
PSFT	**26.89**	89.11	**68.69**	**56.58**	**25.85**	**77.90**	69.75	**59.25**

benchmark: *(i) In-domain tasks:* AIME24, AIME25, AMC, MATH-500 (Hendrycks et al., 2021), OlympidBench (He et al., 2024), and Minerva (Lewkowycz et al., 2022). *(ii) Out-of-domain tasks:* GPQA (Rein et al., 2024), ARC-C (Clark et al., 2018), TruthfulQA (Lin et al., 2021), MMLU-Pro (Wang et al., 2024), SuperGPQA (Du et al., 2025), HeadQA (Vilares & G´omez-Rodr´ıguez, 2019) and IFEval (Zhou et al., 2023). *(3) Inference.* The inference length is set to 10,240 tokens (4,096 for IFEval), with a top-p of 0.95 and a temperature of 0.7.

基准：(i) 域内任务：AIME24、AIME25、AMC、MATH-500（Hendrycks 等，2021）、OlympidBench（He 等，2024）和 Minerva（Lewkowycz 等，2022）。(ii) 域外任务：GPQA（Rein 等，2024）、ARC-C（Clark 等，2018）、TruthfulQA（Lin 等，2021）、MMLU-Pro（Wang 等，2024）、SuperGPQA（Du 等，2025）、HeadQA（Vilares & G´omez-Rodr´ıguez，2019）和 IFEval（Zhou 等，2023）。(3) 推理。推理长度设置为 10,240 个标记（IFEval 为 4,096 个标记），top-p 为 0.95，温度为 0.7。

Analysis 1. PSFT with warm-up consistently outperforms standard SFT on in-domain tasks. Table 1 presents the performance across various in-domain benchmarks, where all fine-tuned methods achieve notable improvements over the original model. PSFT with a warm-up phase not only stabilizes early training dynamics as illustrated in Figure 2, but also enhances robustness across different evaluation benchmarks.

分析 1. 带有预热阶段的 PSFT 在领域内任务中始终优于标准的 SFT。表 1 展示了各种领域内基准测试的性能，其中所有微调方法都显著优于原始模型。带有预热阶段的 PSFT 不仅如图 2 所示稳定了早期的训练动态，还增强了在不同评估基准上的鲁棒性。

Analysis 2: PSFT demonstrates strong generalization. Table 2 shows different performance on out-of-domain benchmarks. Overall, our results are consistent with those reported in Zhou et al. (2025): injecting long CoT data into the model improves its general reasoning ability. However, on IFEval, which evaluates instruction-following capability, SFT, SFT-KL, and PSFT with warm-up significantly degrade the original performance. In contrast, PSFT not only preserves instruc-tion compliance but also substantially enhances the model's reasoning ability across other domains.

分析 2：PSFT 表现出强大的泛化能力。表 2 展示了模型在领域外基准测试中的不同表现。总体而言，我们的结果与周等人（2025）报告的结果一致：将长的 CoT 数据注入模型中可以提高其一般推理能力。然而，在评估指令遵循能力的 IFEval 上，SFT、SFT-KL 和带预热的 PSFT 显著降低了原始性能。相反，PSFT 不仅保持了指令遵循性，还显著增强了模型在其他领域中的推理能力。

PSFT with a warm-up phase also consistently surpasses standard SFT on out-of-domain tasks.

带有预热阶段的 PSFT 在领域外任务中也一直超越标准 SFT。

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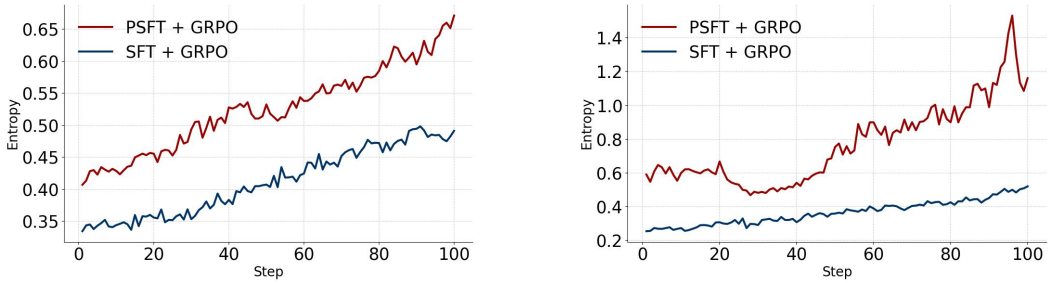
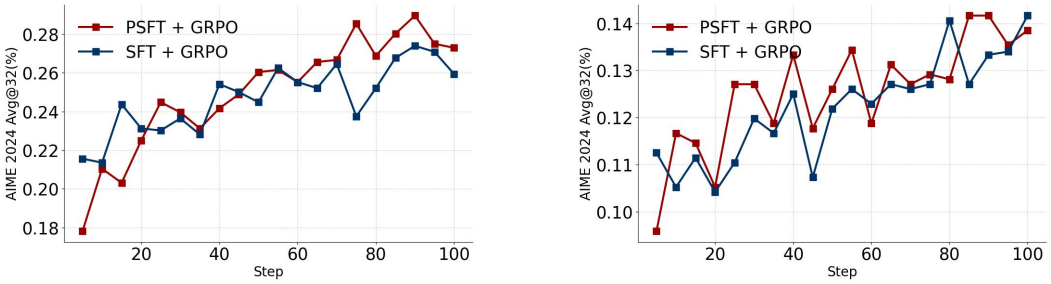


(a) Qwen2.5-7B-Instruct (b) LLama3.1-8B-Instruct

Figure 4: Training dynamics of Entropy on RL experiments.

(a) Qwen2.5-7B-Instruct (b) LLama3.1-8B-Instruct 图 4：在强化学习实验中熵的训练动态。



(a) Qwen2.5-7B-Instruct (b) LLama3.1-8B-Instruct

(a) Qwen2.5-7B-Instruct (b) LLama3.1-8B-Instruct

Figure 5: Training dynamics of in-domain performance on RL experiments.

图 5：在 RL 实验中领域内性能的训练动态。

Analysis 3: PSFT demonstrates robustness across models. For example, when evaluating GPQA and TruthfulQA tasks, SFT and SFT-KL exhibit varying behaviors across different mod-els—one sometimes compromises performance while the other may boost it. In contrast, PSFT consistently improves performance, demonstrating a strong advantage over Llama models.

分析 3：PSFT 在不同模型中表现出鲁棒性。例如，在评估 GPQA 和 TruthfulQA 任务时，SFT 和 SFT-KL 在不同模型中表现出不同的行为——有时会牺牲性能，而有时则能提升性能。相比之下，PSFT 始终能够提升性能，展现出相对于 Llama 模型的强大优势。

4.2 Exploration on the Potential of Models in the RL stage

4.2 在强化学习阶段模型潜力的探索

Practical LLMs often conduct RL after SFT in post-training. The supervised fine-tuned models should function well as an appropriate start point of the following RL stage, avoid both overfitting and underfitting, and better stimulate the potential of RL. In this subsection, we evaluate the power of PSFT in the RL stage.

实用的大型语言模型通常在微调（SFT）之后进行强化学习（RL）。监督微调模型应能作为后续强化学习阶段的良好起点，避免过拟合和欠拟合，并更好地激发强化学习的潜力。在本小节中，我们评估 PSFT 在强化学习阶段中的效果。

Setup. We adopt all techniques from DAPO (Yu et al., 2025), setting clip-higher to 0.28. The RL training uses the DAPO-MATH-17k dataset (Shao et al., 2024), with detailed training configurations provided in Appendix A.1.2. We further select the checkpoints based on the highest in-domain performance for evaluation.

设置。我们采用 DAPO（Yu 等，2025）中的所有技术，将 clip-higher 设置为 0.28。强化学习训练使用 DAPO-MATH-17k 数据集（Shao 等，2024），详细的训练配置见附录 A.1.2。我们进一步根据最高的领域内性能选择检查点进行评估。

4.2.1 Training Dynamics on RL

4.2.1 在强化学习中的训练动态

Analysis 1: PSFT leaves much room for RL optimization. The RL entropy evolution is shown in Figure 4. As low entropy limits the model's exploration, it is notable that using PSFT as a cold start results in higher entropy throughout training, with a steeper increase.

分析 1：PSFT 为 RL 优化留下了很大空间。RL 熵的演变如图 4 所示。由于低熵会限制模型的探索能力，值得注意的是，使用 PSFT 作为冷启动可以在整个训练过程中保持更高的熵，且增加速度更快。

Analysis 2: PSFT shows slow start and rapid catch-up in RL. The in-domain performance evolution is presented in Figure 5. Thanks to its high entropy, which promotes exploration, using PSFT as the cold start leads to a relatively steep performance improvement and ultimately surpasses the performance achieved when using SFT as the cold start.

分析 2：PSFT 在强化学习中表现出缓慢启动和快速追赶的特点。域内性能的演变情况如图 5 所示。得益于其高熵值，能够促进探索，使用 PSFT 作为冷启动可以带来相对较快的性能提升，并最终超越使用 SFT 作为冷启动所达到的性能水平。

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Table 3: Detailed results of **in-domain** performance on RL experiments. For AIME and AMC, the results are avg.@32. For the rest, the results are avg.@8.

表 3：RL 实验中域内性能的详细结果。对于 AIME 和 AMC，结果为 avg.@32。对于其他情况，结果为 avg.@8。

Method	AIME24	AIME25	AMC	MATH-500	OlympidBench	Minerva	Avg.
			Qwen2.5-7B-Instruct				
SFT	22.08	23.02	62.73	84.10	52.35	43.66	47.99
,→GRPO	27.40	26.15	70.86	86.60	58.09	45.27	52.40
PSFT	19.38	21.98	62.34	83.35	51.50	43.33	46.98
,→GRPO	**28.13**	**27.19**	**71.72**	**87.48**	**58.50**	**46.83**	**53.31**
			Llama3.1-8B-Instruct				
SFT	10.63	16.77	47.19	72.60	41.43	32.17	36.80
,→GRPO	14.06	17.50	54.38	77.03	47.13	33.09	40.53
PSFT	10.31	14.48	46.80	71.98	39.61	32.40	35.93
,→GRPO	**14.27**	**18.02**	**55.23**	**77.78**	**47.74**	**37.55**	**41.77**

Table 4: Detailed results of **out-of-domain** performance on RL experiments. For GPQA, ARC-C, TruthfulQA, and IFEval, the results are avg.@8. For the rest, the results are pass.@1.

表 4：强化学习实验中域外性能的详细结果。对于 GPQA、ARC-C、TruthfulQA 和 IFEval，结果为 avg.@8。对于其余项目，结果为 pass.@1。

Method	GPQA	ARC-C	TruthfulQA	MMLU-Pro	SuperGPQA	HeadQA	$\text{IFEval}_{\text{loose}}$	Avg.
				Qwen2.5-7B-Instruct				
SFT	32.89	92.22	63.14	58.98	29.02	74.65	54.42	57.90
,→GRPO	39.39	92.25	61.93	62.61	33.80	75.42	53.89	59.90
PSFT	33.21	92.29	**67.16**	59.18	28.10	75.82	73.03	61.26
,→GRPO	**43.43**	**92.42**	64.84	**63.65**	**34.51**	**75.89**	**73.73**	**64.06**
				Llama3.1-8B-Instruct				
SFT	19.38	87.73	67.08	50.29	21.95	73.52	33.45	50.49
,→GRPO	31.06	88.57	64.82	54.39	27.80	73.81	31.85	53.19
PSFT	26.89	89.11	**68.69**	56.58	25.85	77.90	69.75	59.25
,→GRPO	**36.23**	**90.92**	65.24	**60.92**	**31.30**	**78.12**	**71.02**	**61.96**

4.2.2 Detailed Evaluation

Analysis 3: PSFT achieves better in-domain performance after RL. Table 3 presents the results of RL experiments on in-domain benchmarks. Overall, RL improves the performance of all models. A comprehensive evaluation of the in-domain tasks shows that, although PSFT initially lags behind SFT, it holds significant potential that can be further unlocked through RL.

领域内基准的强化学习（RL）实验。总体而言，强化学习提高了所有模型的性能。对领域内任务的全面评估表明，尽管 PSFT 最初落后于 SFT，但其具有巨大的潜力，可以通过强化学习进一步挖掘。

Analysis 4: PSFT outperforms SFT on out-of-domain RL tasks by a large margin. Table 4 presents the results of RL experiments on out-of-domain benchmarks. It is evident that if the original model exhibits weak capabilities during the cold-start phase, the subsequent RL training is constrained by these limitations. For instance, the IFEval results very clearly reflect this point. This further underscores the necessity of enhancing the current SFT algorithm.

分析 4：PSFT 在领域外的强化学习任务中表现显著优于 SFT。表 4 展示了在领域外基准上的强化学习实验结果。很明显，如果原始模型在冷启动阶段表现出较弱的能力，后续的强化学习训练就会受到这些限制的影响。例如，IFEval 的结果非常清楚地反映了这一点。这进一步强调了改进当前 SFT 算法的必要性。

4.3 Further Experiments on Human Alignment

4.3 关于人类对齐的进一步实验

In this section, to demonstrate the universality of PSFT, we extend our experiments to human alignment datasets, employing different base models and algorithms such as DPO (Rafailov et al., 2023). Our results show that PSFT effectively reduces the alignment tax (i.e., the generalization gap) while still leaving room for further optimization.

在本节中，为了展示 PSFT 的普适性，我们将实验扩展到人类对齐数据集，采用不同的基础模型和算法，如 DPO（Rafailov 等，2023）。我们的结果表明，PSFT 有效地减少了对齐成本（即泛化差距），同时仍为进一步优化留下了空间。

Setup. We adopt a completely different setup to demonstrate the universality of PSFT further. *(1) Model and Dataset.* We fine-tune the pre-trained Qwen3-4B-Base (Yang et al., 2025) model with the UltraFeedback dataset (Cui et al., 2023). *(2) Training algorithm.* We select the chosen region of the dataset and the SFT/PSFT model on it. And for PSFT, we train a double step named $\text{PSFT}_{\text{prolong}}$. Then, use the DPO algorithm to enable the model to learn the contrastive reward signal.

设置。我们采用完全不同的设置来进一步展示 PSFT 的普适性。(1) 模型和数据集。我们使用 UltraFeedback 数据集（Cui 等，2023）对预训练的 Qwen3-4B-Base（Yang 等，2025）模型进行微调。(2) 训练算法。我们选择数据集中的选定区域以及其上的 SFT/PSFT 模型。对于 PSFT，我们训练一种名为 PSFTprolong 的双步方法。然后，使用 DPO 算法使模型学习对比奖励信号。

Proximal Supervised Fine-Tuning

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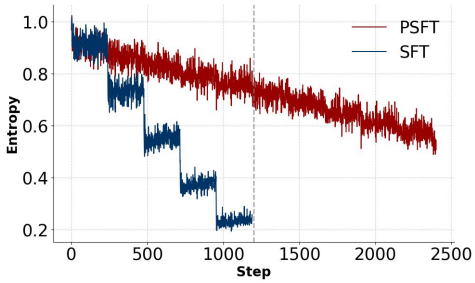
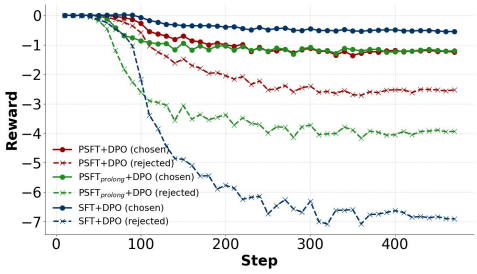


(a) Entropy evolution during SFT/PSFT training.

(a) 熵在 SFT/PSFT 训练中的演变。



(b) Positive/Negative reward progression during DPO training.

(b) 正/负奖励在 DPO 训练中的变化趋势。

Figure 6: Training dynamics of SFT/PSFT followed by DPO

图 6：SFT/PSFT 之后的 DPO 训练动态

Table 5: Qwen3-4B-Base DPO training on the alignment benchmarks.

表 5：Qwen3-4B-Base 在对齐基准上的 DPO 训练。

Method	AlpacaEval2		Arena-Hard	MT-Bench	
	LC (%)	WR (%)	WR (%)	1-turn	2-turn
SFT	12.24	8.52	17.90	7.64	5.71
,→DPO	16.96	13.40	26.50	7.91	6.00
PSFT$_{prolong}$	11.95	8.37	17.40	7.41	5.84
,→DPO	19.26	15.17	30.20	7.63	6.74
PSFT	11.79	7.49	11.40	6.83	4.86
,→DPO	**23.29**	**20.13**	**36.40**	**8.51**	**6.95**

(3) Evaluation. Since short CoT models can not solve the complex problems, we use the ARC, GSM8K (Cobbe et al., 2021), MMLU (Hendrycks et al., 2020), GPQA, and TruthfulQA to evaluate the alignment tax. We evaluate our models on three alignment benchmarks: MT-Bench (Zheng et al., 2024a), AlpacaEval (Dubois et al., 2024), and Arena-Hard (Li et al., 2024a). We use Qwen3-30B-A3-Instruct-2507 (Yang et al., 2025) as the judge model to provide alignment evaluation. We use llm-eval-harness (Gao et al., 2024) to evaluate the alignment tax performance. More experimental setup, see Appendix A.1.3.

（3）评估。由于短程推理模型无法解决复杂问题，我们使用 ARC、GSM8K（Cobbe 等，2021）、MMLU（Hendrycks 等，2020）、GPQA 和 TruthfulQA 来评估对齐税。我们在三个对齐基准测试上评估我们的模型：MT-Bench（Zheng 等，2024a）、AlpacaEval（Dubois 等，2024）和 Arena-Hard（Li 等，2024a）。我们使用 Qwen3-30B-A3-Instruct-2507（Yang 等，2025）作为判断模型，以提供对齐评估。我们使用 llm-eval-harness（Gao 等，2024）来评估对齐税的性能。更多实验设置，请参见附录 A.1.3。

4.3.1 Training Dynamics

4.3.1 训练动态

Analysis 1: PSFT still reliably prevents entropy collapse for other domains. As shown in Figure 6a, the entropy loss of SFT still shows a severe sawtooth shape, indicating a potential overfitting phenomenon, while PSFT overcomes this point. Therefore, SFT leads to a severe alignment tax (Figure 7) and restricts further optimization (Table 5).

分析 1：PSFT 仍然可靠地防止了其他领域中的熵崩溃。如图 6a 所示，SFT 的熵损失仍然表现出严重的锯齿形状，表明可能存在过拟合现象，而 PSFT 克服了这一点。因此，SFT 导致严重的对齐代价（图 7），并限制了进一步的优化（表 5）。

4.3.2 Human Alignment Performance

4.3.2 人类对齐性能

Analysis2. Table 5 presents the results of DPO experiments on alignment benchmarks. We observe that $\text{PSFT}_{\text{prolong}}$ achieves a similar effect to SFT training while avoiding entropy collapse, and it performs even better during DPO training. Moreover, using PSFT as the cold-start provides additional benefits that are further unlocked by DPO. This is consistent as reported by Xiao (2024) that when an SFT model collapses—i.e., becomes overly biased toward certain outputs with near certainty (Figure 6b)—it can subsequently induce preference collapse in the alignment process.

分析 2。表 5 展示了在对齐基准测试中进行 DPO 实验的结果。我们观察到，PSFTprolong 在避免熵坍缩的情况下，实现了与 SFT 训练相似的效果，并且在 DPO 训练期间表现得更好。此外，使用 PSFT 作为冷启动提供了额外的优势，这些优势通过 DPO 进一步解锁。这与小（2024）报告的一致，即当一个 SFT 模型发生坍缩——也就是说，过于偏向某些输出（几乎确定）时（图 6b）——它随后可能在对齐过程中引发偏好坍缩。

Analysis 3: PSFT training exploits both positive and negative samples well. As shown in Figure 6b, using SFT or PSFT as the cold-start point can influence the subsequent DPO phase. $\text{PSFT}_{\text{prolong}}$ and PSFT yield similar rewards on positive samples; the main difference is that pro-longed training reduces the occurrence of negative samples. In contrast, SFT training results in positive samples appearing frequently while negative samples are rare, which limits the model's ability to learn from the reward signal and instead biases it toward the human-selected data.

分析 3：PSFT 训练很好地利用了正样本和负样本。如图 6b 所示，以 SFT 或 PSFT 作为冷启动点会影响后续的 DPO 阶段。PSFTprolong 和 PSFT 在正样本上产生相似的奖励；主要区别在于延长训练减少了负样本的出现。相比之下，SFT 训练导致正样本频繁出现而负样本很少，这限制了模型从奖励信号中学习的能力，并使其偏向于人类选择的数据。

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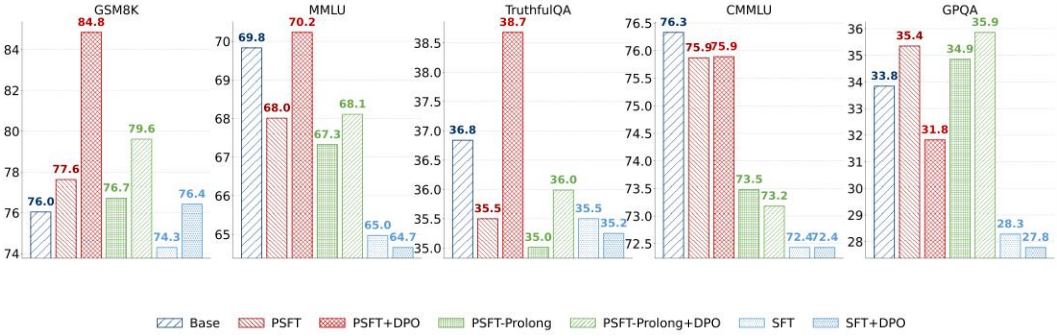
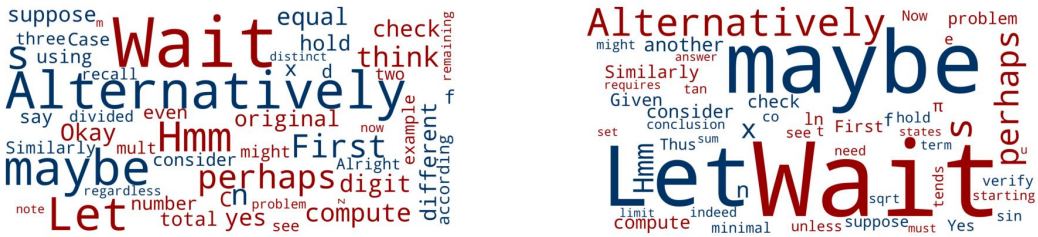


Figure 7: Results of models on alignment tax (out-of-domain tasks).

图 7：模型在对齐税（领域外任务）上的结果。



(a) Clipped token cloud in epoch 1.

(b) Clipped token cloud in epoch 3.

(a) 第 1 个周期的截断标记云。(b) 第 3 个周期的截断标记云。

Figure 8: Examples and Changes of clipped tokens in PSFT during training.

图 8：在训练过程中 PSFT 中截断标记的示例和变化。

Analysis4. Figure 7 shows the alignment tax results with various tasks (e.g., GSM8K, MMLU, GPQA) as the out-of-domain tasks. It indicates that: (a) PSFT and PSFT+DPO could largely maintain the general ability during the SFT stage compared to the original SFT. (b) PSFT-Prolong, even being degraded by possible overfitting issues, achieves relatively good results on alignment tax, implying the robustness of PSFT in practical scenarios.

分析 4。图 7 展示了在不同任务（例如 GSM8K、MMLU、GPQA）作为域外任务时的对齐税结果。这表明：(a) 与原始 SFT 相比，PSFT 和 PSFT+DPO 在 SFT 阶段能够很大程度上保持一般能力。(b) PSFT-Prolong 即使可能因过拟合问题而性能下降，但在对齐税上仍取得了相对较好的结果，这表明 PSFT 在实际场景中的鲁棒性。

5 In-Depth Analyses

5 深入分析

5.1 Clipped Tokens in PSFT

5.1 PSFT 中的截断标记

Figure 8 shows the representative clipped tokens during our PSFT's training process. As we can see, the clipped token mostly concentrates on uncertain words like "wait", "alternatively", and similar words that reflect certain "long thinking patterns", which are also supposed to be learned in the following RL stage. With the training in progress, these token-clip weights are more pronounced, while other tokens become smaller. The "thinking pattern" has been gradually and smoothly learned into models via PSFT, without much disturbance to the general capabilities.

图 8 展示了我们在 PSFT 训练过程中出现的代表性裁剪标记。如我们所见，这些被裁剪的标记主要集中在像"wait"（等待）、"alternatively"（或者）等不确定的词语上，这些词语反映了某种"长时间思考模式"，这些模式也应在随后的强化学习阶段中进行学习。随着训练的进行，这些标记裁剪权重变得更加明显，而其他标记的权重则逐渐减小。通过 PSFT，这些"思考模式"已被逐步且平滑地融入模型中，而对模型的一般能力影响不大。

5.2 Parameter Analysis on Clipped Value

5.2 截断值的参数分析

As shown in Figure 9, PSFT without clipping maintains higher entropy but suffers from large, unstable gradient norms and highly fluctuating downstream results. By introducing the trust-region clipping mechanism, PSFT strikes a better balance: it prevents entropy collapse, stabilizes gradient updates, and achieves steady improvements in performance. Across different clipping thresholds, with medium values of ϵ (e.g., 0.28) showing particularly favorable trade-offs between stability and performance.

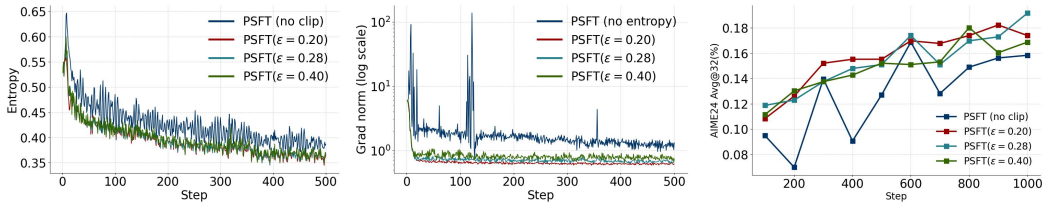
如图 9 所示，未使用截断的 PSFT 保持了较高的熵，但存在较大的、不稳定的梯度范数和下游结果的剧烈波动。通过引入信任区域截断机制，PSFT 取得了更好的平衡：它防止了熵坍缩，稳定了梯度更新，并实现了性能的持续提升。在不同的截断阈值下，采用中等值的 ε（例如 0.28）在稳定性和性能之间表现出特别有利的权衡。

6 Related Work

6 相关工作

Supervised Fine-tuning. SFT is typically the initial step in the post-training pipeline. Addressing potential issues at the SFT stage can significantly enhance the robustness of the resulting further fine-tuned model. Several studies have noted that the standard cross-entropy (CE) loss may not be

监督微调。SFT 通常是训练后的初始步骤。在 SFT 阶段解决潜在问题可以显著提高最终微调模型的鲁棒性。一些研究指出，标准的交叉熵（CE）损失可能并不



(a) Entropy (b) Gradient Norm (c) Performance

Figure 9: In-domain results of PSFT with different clipped values.

图 9：PSFT 在不同截断值下的领域内结果。

the most suitable objective for SFT (Li et al., 2024b; Xiao, 2024), arguing that training with CE at this stage tends to encourage the model to memorize the training data rather than acquire more generalizable capabilities. Li et al. (2024b) proposes the GEM, a game-theoretic SFT algorithm with entropy regularization to preserve diversity and mitigate forgetting. The concurrent work is Importance-weighted SFT (iw-SFT) (Qin & Springenberg, 2025) and DFT Wu et al. (2025). The key insight of iw-SFT is to reinterpret standard SFT as optimizing a loose lower bound on the RL objective in a sparse reward setting. While vanilla SFT simply maximizes the likelihood of filtered"successful" trajectories, this lower bound becomes increasingly loose as the model distribution drifts away from the reference policy. iw-SFT addresses this gap by introducing importance re-weighting, which assigns higher weights to more preferred trajectories and thus tightens the bound toward the true RL objective. DFT views SFT as a flawed policy gradient and introduces a simple probability-based reweighting that improves generalization.

SFT（Li 等，2024b；Xiao，2024）中最合适的目标，认为在这个阶段使用 CE 进行训练往往会促使模型记忆训练数据，而不是获得更普遍适用的能力。Li 等（2024b）提出了 GEM，这是一种基于博弈论的 SFT 算法，通过熵正则化来保持多样性并减少遗忘。同时进行的工作包括基于重要性加权的 SFT（iw-SFT）（Qin & Springenberg，2025）和 DFT（Wu 等，2025）。iw-SFT 的关键洞察在于重新解释标准的 SFT 为在稀疏奖励设置下优化 RL 目标的一个松散下界。虽然普通的 SFT 只是最大化过滤后的“成功”轨迹的似然，但随着模型分布远离参考策略，这个下界会变得越来越松散。iw-SFT 通过引入重要性重加权来解决这一差距，该方法赋予更受青睐的轨迹更高的权重，从而将下界向真实的 RL 目标收紧。DFT 将 SFT 视为有缺陷的策略梯度，并引入了一种简单的基于概率的重加权方法，以提高泛化能力。

Reinforcement Learning. Following SFT, RL is employed to optimize reward signals directly. Policy gradient methods (Sutton et al., 2000) provide the foundation but are prone to instability and high variance. To address this, TRPO (Schulman et al., 2015) introduces a trust-region constraint, limiting each policy update to a small KL-divergence neighborhood to ensure stable improvement. PPO (Schulman et al., 2017) further simplifies this approach with a clipped surrogate objective, striking a balance between stability and efficiency.

强化学习。在 SFT 之后，RL 被用来直接优化奖励信号。策略梯度方法（Sutton 等，2000）提供了基础，但容易出现不稳定和高方差的问题。为了解决这个问题，TRPO（Schulman 等，2015）引入了一个信任区域约束，将每个策略更新限制在小的 KL 散度邻域内，以确保稳定的改进。PPO（Schulman 等，2017）则通过一个裁剪的代理目标进一步简化了这种方法，在稳定性和效率之间取得了平衡。

7 CONCLUSION

7 结论

Motivated by TRPO and PPO, we propose Proximal Supervised Fine-Tuning (PSFT). PSFT preserves a smooth entropy curve while achieving performance comparable to standard SFT at similar entropy levels, and it exhibits strong generalization. Furthermore, using PSFT as a cold-start model facilitates more effective post-training techniques such as RL optimization and DPO. Overall, PSFT presents a promising alternative to traditional SFT.

受 TRPO 和 PPO 的启发，我们提出了近端监督微调（PSFT）。PSFT 在保持平滑熵曲线的同时，在相似熵水平下实现了与标准 SFT 相当的性能，并表现出强大的泛化能力。此外，使用 PSFT 作为冷启动模型有助于更有效地应用后续训练技术，如强化学习优化和 DPO。总体而言，PSFT 为传统的 SFT 提供了一个有前景的替代方案。

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A APPENDIX

附录

A.1 EXPERIMENTAL DETAILS

A.1 实验细节

We perform SFT, PSFT, and RL training using the verl framework (Sheng et al., 2024), and employ LLama-Factory (Zheng et al., 2024b) for DPO training. The loss is aggregated using token-mean in verl. For SFT and PSFT, we use a weight decay of 0.1. All experiments are conducted with full fine-tuning.

我们使用 verl 框架（Sheng 等，2024）进行 SFT、PSFT 和 RL 训练，并使用 LLama-Factory（Zheng 等，2024b）进行 DPO 训练。在 verl 中，损失函数使用 token-mean 进行聚合。对于 SFT 和 PSFT，我们采用 0.1 的权重衰减。所有实验均采用全量微调进行。

A.1.1 MATH REASONING EXPERIMENTS

数学推理实验

Method	Train batch size	Mini batch size	Learning rate	Train epochs	High clip	Cutofflen
PSFT	256	32	1e-6	10	0.28	10k
SFT	256	–	2e-5	10	–	10k

Table 6: PSFT/SFT experiment configuration

表 6：PSFT/SFT 实验配置

The detailed training configurations for SFT/PSFT are presented in Table 6.

SFT/PSFT 的详细训练配置见表 6。

A.1.2 EXPLORATION ON MODEL POTENTIAL IN RL

A.1.2 在强化学习中模型势能的探索

The detailed training configurations for RL are presented in Table 7. For Qwen models and LLama models, we use the same configuration.

强化学习的详细训练配置如表 7 所示。对于 Qwen 模型和 LLama 模型，我们使用相同的配置。

Config SFT + GRPO PSFT + GRPO 配置 SFT + GRPO PSFT + GRPO		
lr	1e-6	1e-6
klcoef	0.0	0.0
maxpromptlength	2k	2k
maxresponselength	10k	10k
overlongbuffer.len	2k	2k
trainbatchsize	256	256
ppominibatchsize	32	32
clipratiolow	0.2	0.2
clipratiohigh	0.28	0.28
temperature	1.0	1.0
rollout.n	8	8
totaltrainingsteps	100	100

Table 7: RL experiment configuration
表 7：强化学习实验配置

Method 方法	Train batch size 训练批次大小	Mini batch size 小批次大小	Learning rate 学习率	Train epochs 训练轮数	High clip 高剪辑	Cutoff len 截断长度
PSFT	256	32	1e-6	5	0.28	6k
$PSFT_{prolong}$	256	32	1e-6	10	0.28	6k
SFT	256	–	2e-5	5	–	6k

Table 8: PSFT/SFT experiment configuration
表 8：PSFT/SFT 实验配置

Method	Train batch size	β	Learning rate	Train epochs	Cutoff len
DPO	64	0.01	5e-7	1	4k

Table 9: DPO experiment configuration

方法	训练批次大小	β	学习率	训练轮次	截断长度
DPO	64	0.01	5e-7	1	4k

表 9：DPO 实验配置

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A.1.3 ALIGNMENT EXPERIMENTS

A.1.3 对齐实验

Training. The detailed training configurations for SFT/PSFT on alignment tasks are presented in Table 8. The detailed training configurations for DPO are presented in Table 9.

训练。SFT/PSFT 在对齐任务上的详细训练配置见表 8。DPO 的详细训练配置见表 9。

Config	MT-Bench	Alpaca-Eval	Arena-Hard
temperature	0.0	0.7	1.0
topp	–	0.95	0.7

Table 10: Alignment benchmark generation

表 10：对齐基准生成

Evaluation. We adopt the settings listed in Table 10 for evaluation generation. To assess alignment tax, we use the corresponding tasks in llm-eval-harness, as summarized in Table 11.

评估。我们采用表 10 中列出的设置进行评估生成。为了评估对齐税，我们使用 llm-eval-harness 中的相应任务，如表 11 所示。

task	GSM8K	MMLU	TruthfulQA	CMMLU	GPQA
任务	GSM8K	MMLU	TruthfulQA	CMMLU	GPQA
setting	gsm8k cot	mmlu flan n shot	generative	truthfulqa mc1	cmmlu cot n shot
设置	GSM8K COT	MMLU Flan N shot	生成性	TruthfulQA MC1	CMMLU COT N shot

Table 11: Alignment tax evaluation using llm-eval-harness

表 11：使用 llm-eval-harness 进行对齐税收评估

A.2 SYSTEM PROMPT

A.2 系统提示

For the reasoning task, we use the system prompt as follows: Please reason step by step, and put your final answer within \boxed{}.

对于推理任务，我们使用以下系统提示：请逐步推理，并将最终答案放在\boxed{}中。

For the alignment task, we use the system prompt as follows: You are a helpful assistant.

对于对齐任务，我们使用以下系统提示：你是一个有帮助的助手。